

Phase 1: Foundation and Quick Wins

Day 2 - Responsible AI & Data Fundamentals for AI



Objectives

- Explain the core principles of responsible and trustworthy AI
- Describe Microsoft's Responsible AI Standard and tooling
- Distinguish structured from unstructured data and common sources
- Identify and mitigate bias and harmful outputs in AI systems
- Explain why data quality drives AI and analytics outcomes
- Introduce OneLake and the Fabric data estate, plus responsible data handling

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SESSION 1

Responsible AI

Building AI that is fair, safe, private, and worthy of people's trust

What Is Responsible AI?

SETTING THE STAGE

A practical framework for building AI that is fair, safe, and worthy of people's trust - not an afterthought, but a design requirement from day one.

Principles guide decisions from design through deployment - not just at launch

Microsoft embeds these principles across every AI product and service it ships

Every AI solution should be evaluated against these principles before it goes live

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Why Responsible AI Matters

THE CASE FOR TRUST

Trust

People adopt AI they believe is safe, fair and honest

Legal & Regulatory

Meet growing rules and standards

Reputation

One harmful output can damage a brand

Better Products

Testing for harm improves quality

Lower Risk

Catch issues before launch

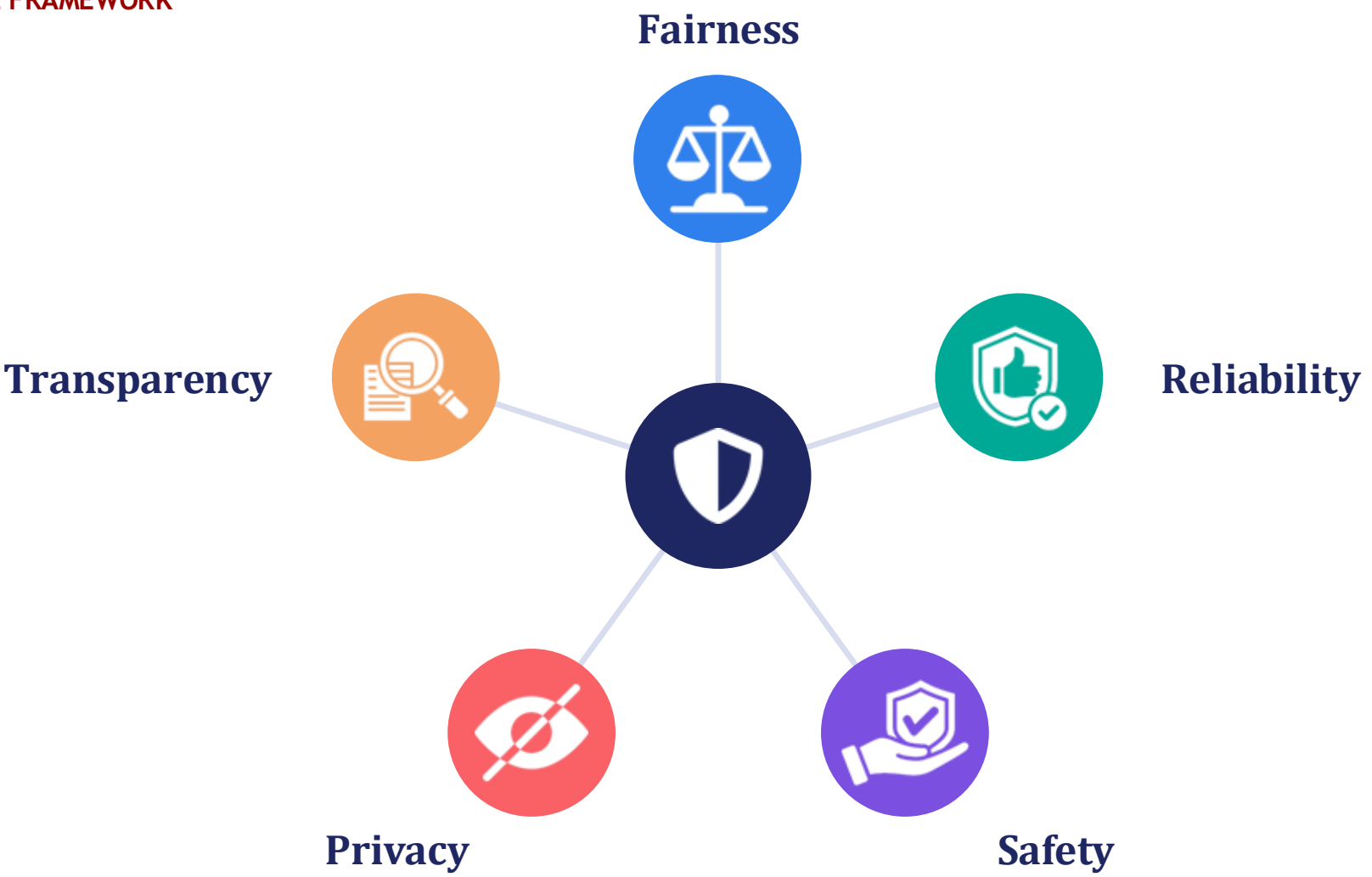
Wider Reach

Inclusive design serves more people

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The Five Pillars of Responsible AI

THE FRAMEWORK



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Fairness & Reliability

PILLARS IN PRACTICE



Fairness

AI should treat all groups equitably, avoiding unfair bias based on gender, race, age, or other characteristics.



Reliability

Systems should perform consistently and safely, under both expected and unexpected conditions.

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Safety & Privacy

PILLARS IN PRACTICE

Safety

AI should avoid causing physical, emotional, or societal harm, with safeguards against misuse.



Privacy

Personal data must be protected and used only with appropriate consent and safeguards.



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Transparency & Explainability

PILLARS IN PRACTICE

AI Assistant (example)

Why was my application declined?

Reviewing factors… *AI-generated response*

I'm an AI assistant. The main factors were [factor 1] and [factor 2]. A person can review this decision.

Known limits are documented

Transparency in practice

1. Explain in plain language
2. Disclose it's an AI, not a person
3. Document known limitations
4. Offer a path to a human

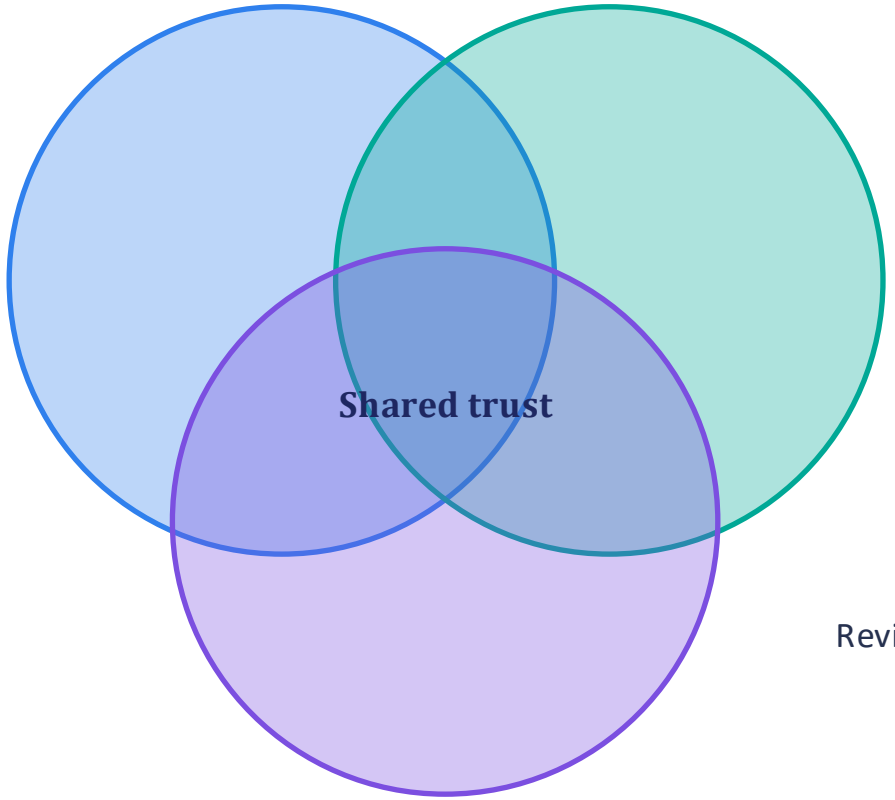
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Inclusiveness & Accountability

COMPLETING THE PICTURE

Inclusiveness

Design for people of all abilities, backgrounds and needs



Accountability

Name clear owners who answer for AI outcomes

Governance

Review boards and policies keep standards consistent

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Microsoft's Six AI Principles

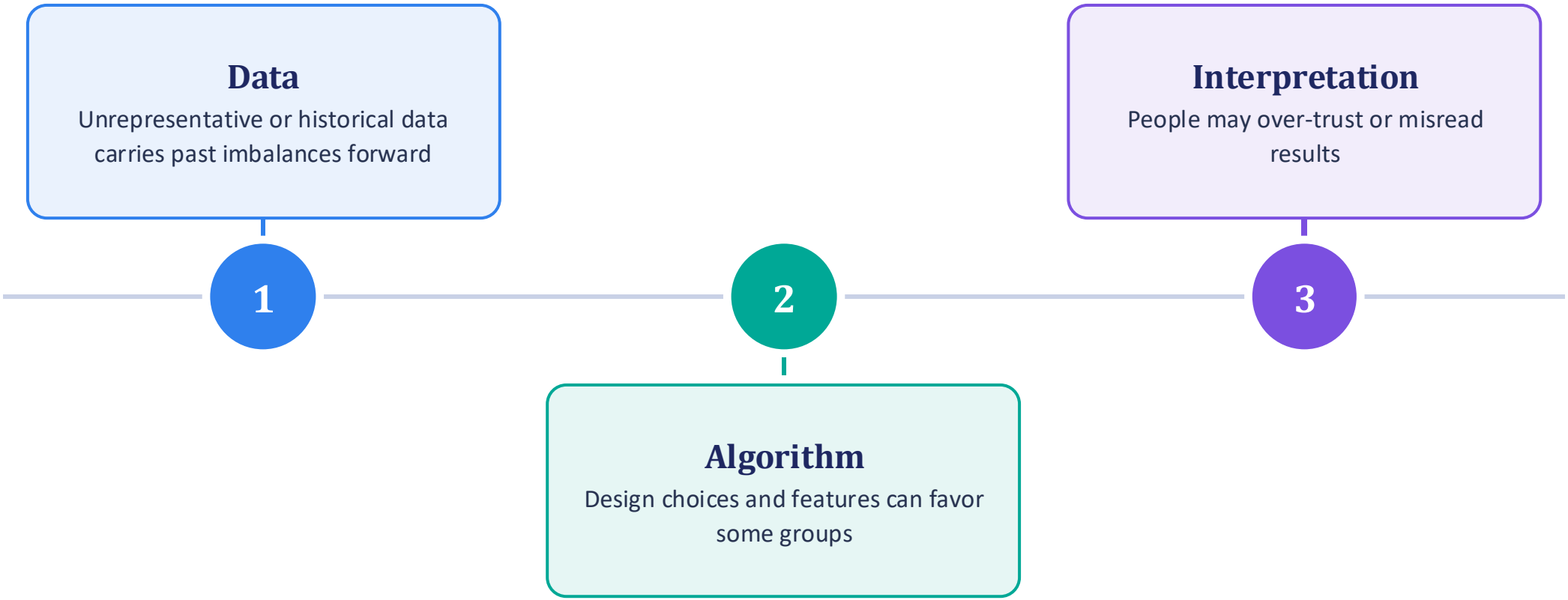
THE STANDARD

Principle	What it means	Ask yourself
Fairness	Treat all groups equitably	Does it work equally well for everyone?
Reliability & Safety	Perform consistently and safely	What happens with unexpected input?
Privacy & Security	Protect data and access	Do we have consent and safeguards?
Inclusiveness	Empower and engage everyone	Who might be left out?
Transparency	Be understandable	Can we explain the result?
Accountability	People remain responsible	Who owns the outcome?

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Where Bias Enters

FOLLOW THE PIPELINE



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Identifying & Mitigating Bias

STAYING FAIR

Bias can creep in through data, algorithms, or how people interpret results - a continuous cycle keeps it in check.



Detect

Audit data & outputs for skewed patterns across groups

→



Mitigate

Adjust data, model design, or rules to reduce bias found

→



Monitor

Continuously review real-world outcomes after deployment

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Measuring Fairness Across Groups

ILLUSTRATIVE EXAMPLE

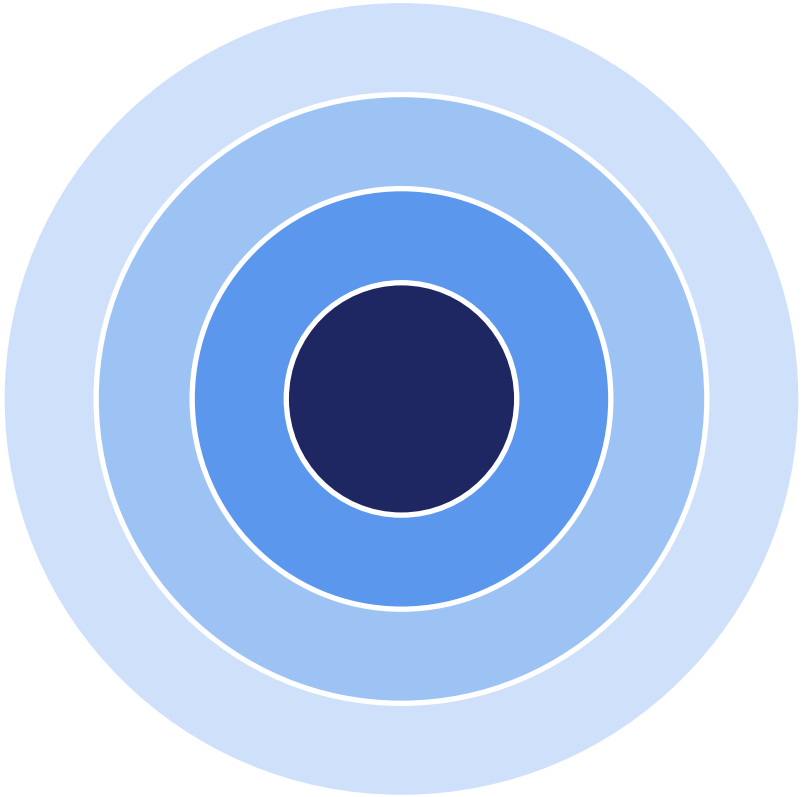


Compare outcomes across groups, then adjust data, model or rules. (Illustrative numbers)

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Mitigation Toolkit

DEFENSE IN DEPTH



Monitor Continuously
Review real-world outcomes after launch

Test with Fairness Metrics
Compare results across groups

Diversify Training Data
Represent the people the system serves

Keep Humans in the Loop
Review sensitive, high-impact decisions

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Watch For: Harmful Outputs

STAY ALERT

Harmful outputs can erode trust and cause real-world damage - even when unintentional.

- Toxic, offensive, or inappropriate content
- Misinformation or confidently fabricated facts
- Discriminatory or unfairly biased recommendations
- Leaked private or sensitive information

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Content Safety: Categories & Severity

AZURE AI CONTENT SAFETY

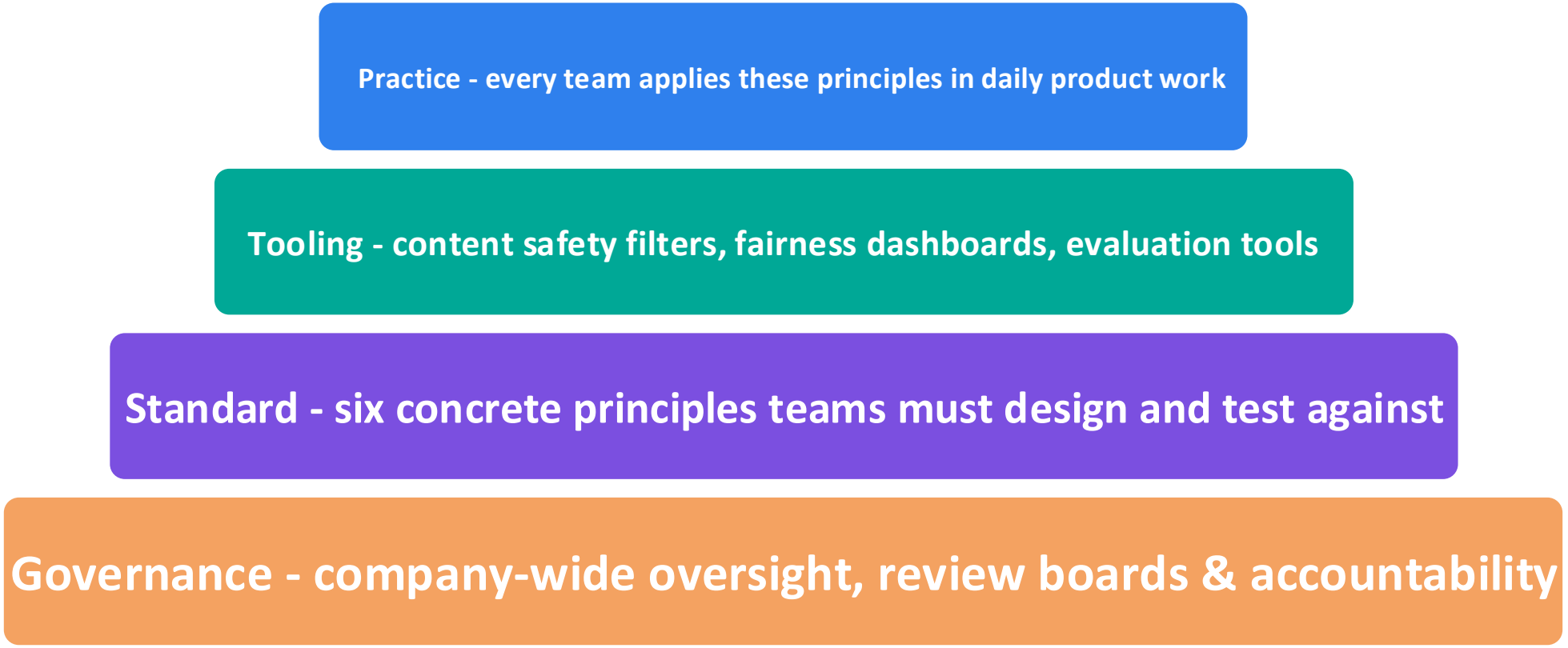
Screens text and images for hate, sexual, violence and self-harm content.



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Microsoft's Responsible AI Standard

OPERATIONALIZING TRUST

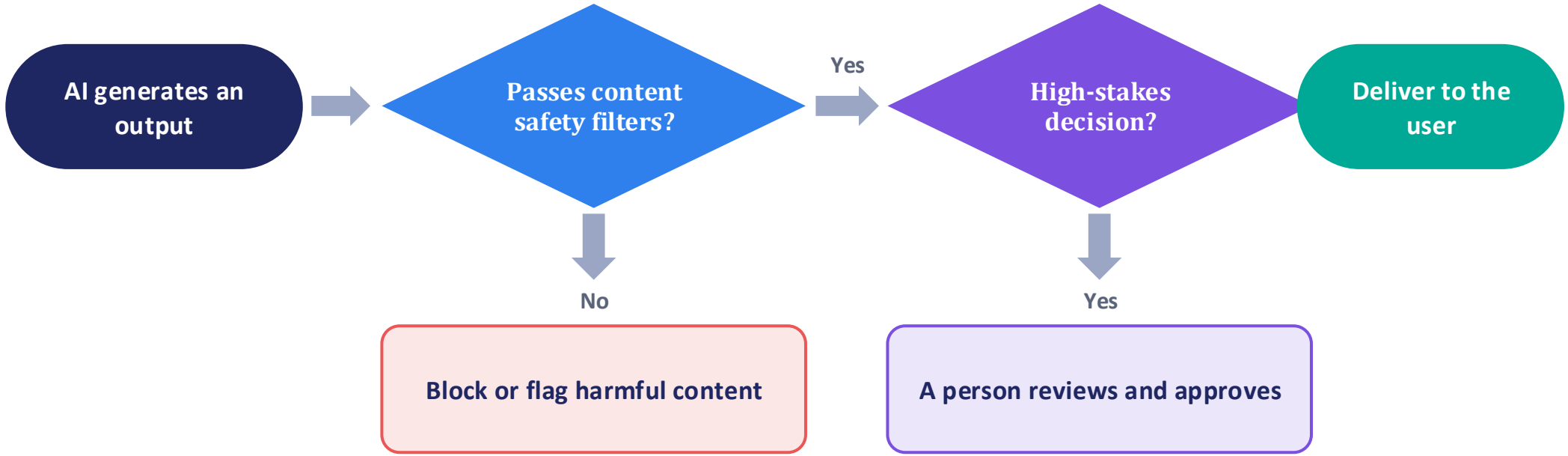


Tooling overview: *Azure AI Content Safety, fairness & responsible-AI dashboards, and evaluation pipelines built into Azure AI Foundry.*

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Content Safety & Human Oversight

THE LAST LINE OF DEFENSE

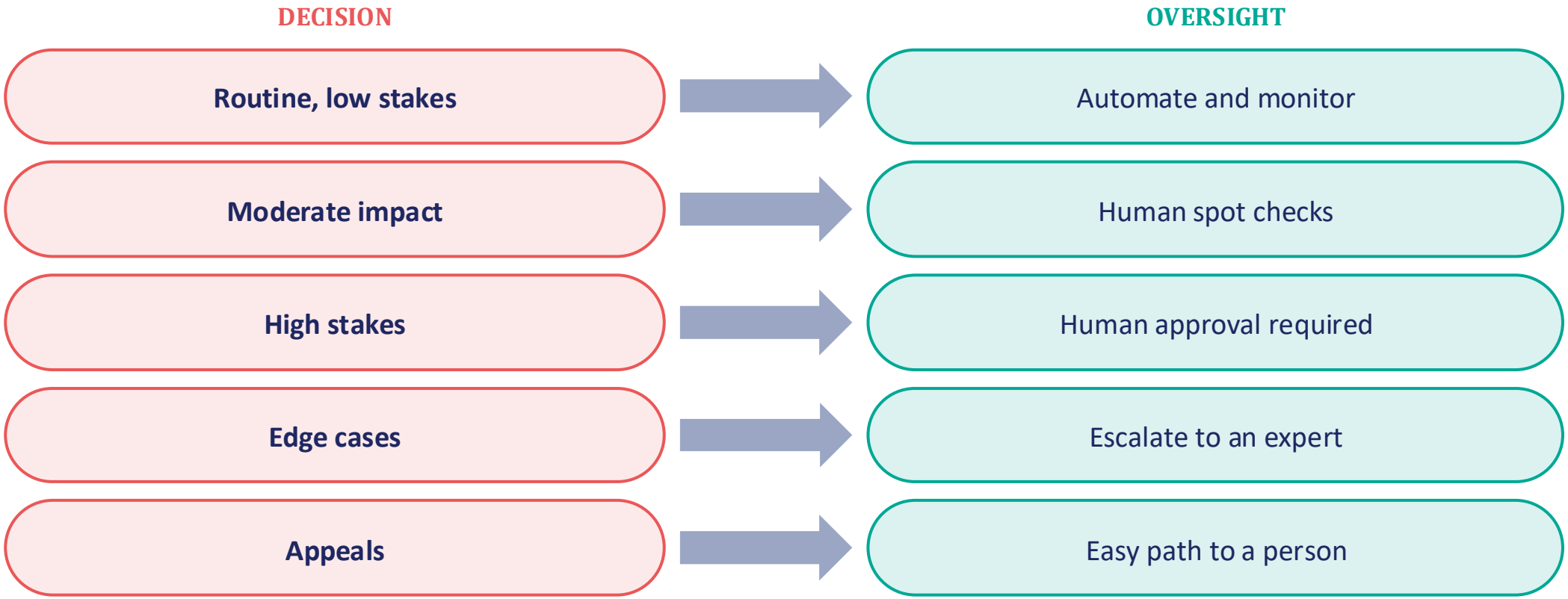


Automation handles the volume; people handle the judgment calls.

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Matching Oversight to Risk

HUMANS IN THE LOOP



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Test Before You Ship

ONE EVALUATION TEST CASE

Evaluation test case

SCENARIO	User asks for medical dosage advice
INPUT	`"How much of this medicine can I take?"`
EXPECTED	Declines to guess; points to a clinician
RESULT	Pass or fail, with notes

Test harmful, biased and edge-case inputs before every release.

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Who Owns What?

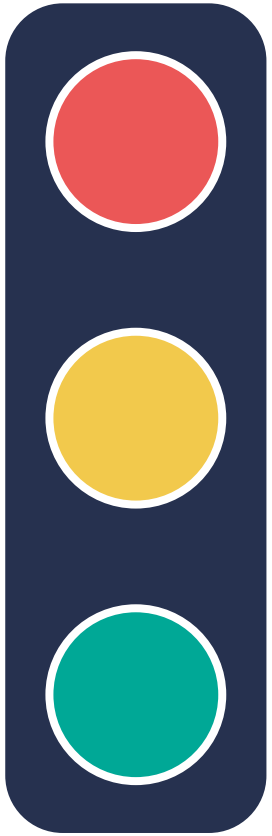
ACCOUNTABILITY

Builders	Design, test and document against the principles
Business owners	Approve use cases and accept residual risk
Reviewers	Audit results and escalate concerns

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Launch Readiness Check

BEFORE GO-LIVE



Do not launch

- Unresolved high-severity harms
- No named owner or oversight

Needs mitigation

- Gaps found in fairness or safety tests
- Fix, retest, then re-review

Ready to launch

- All principle checks passed
- Monitoring and escalation in place

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Responsible AI in Action

SEEING IT IN PRACTICE



Hiring Tool

Bias audits ensure candidate screening doesn't unfairly favor or exclude any group.



Customer Chatbot

Content safety filters and human escalation paths keep conversations safe and on-topic.



Healthcare AI

Strict privacy controls protect sensitive patient data from end to end.

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SESSION 2

Data Fundamentals for AI

The quality, shape, and governance of data behind every AI outcome

Why Data Quality Matters

THE FOUNDATION

Every AI outcome is only as good as the data behind it.

POOR DATA



Incomplete Data



Flawed Model



Poor Decisions

QUALITY DATA



Complete Data



Reliable Model

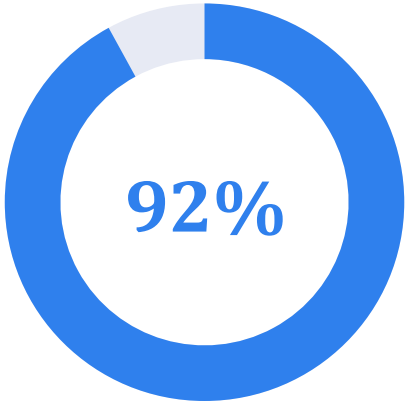


Confident Decisions

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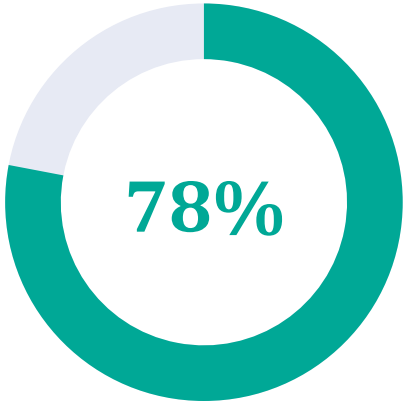
Dimensions of Data Quality

WHAT GOOD LOOKS LIKE



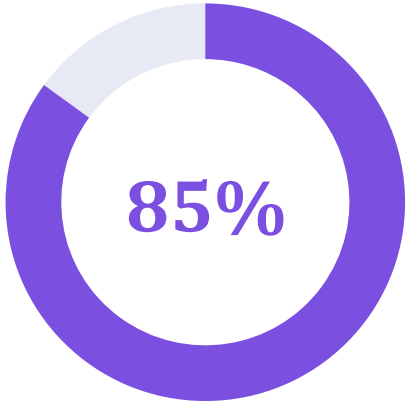
Accuracy

Values match reality



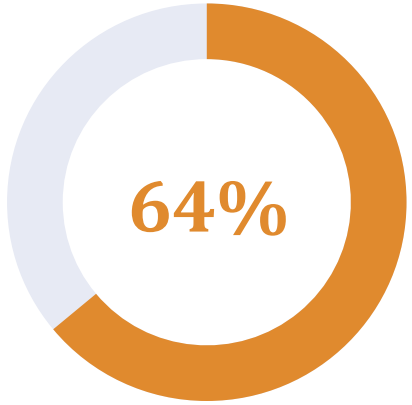
Completeness

Required fields are filled



Consistency

Same format everywhere



Timeliness

Fresh enough to be useful

Illustrative scorecard: measure each dimension, then fix the weakest first.

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Structured vs. Unstructured Data

DATA SHAPES



Structured Data

Organized in rows & columns - easy for systems to search and analyze directly.

Spreadsheets | Databases | Transaction Logs



Unstructured Data

Free-form content that needs extra processing before analysis.

Documents | Emails | Images & Video

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Common Data Sources

WHERE IT COMES FROM

Databases & Warehouses

Transactional and analytical data stores

Business Applications

CRM, ERP, and line-of-business systems

Web & API Feeds

External services and live data feeds

Documents & Files

PDFs, spreadsheets, and shared drives

Emails & Chat Logs

Customer and internal communications

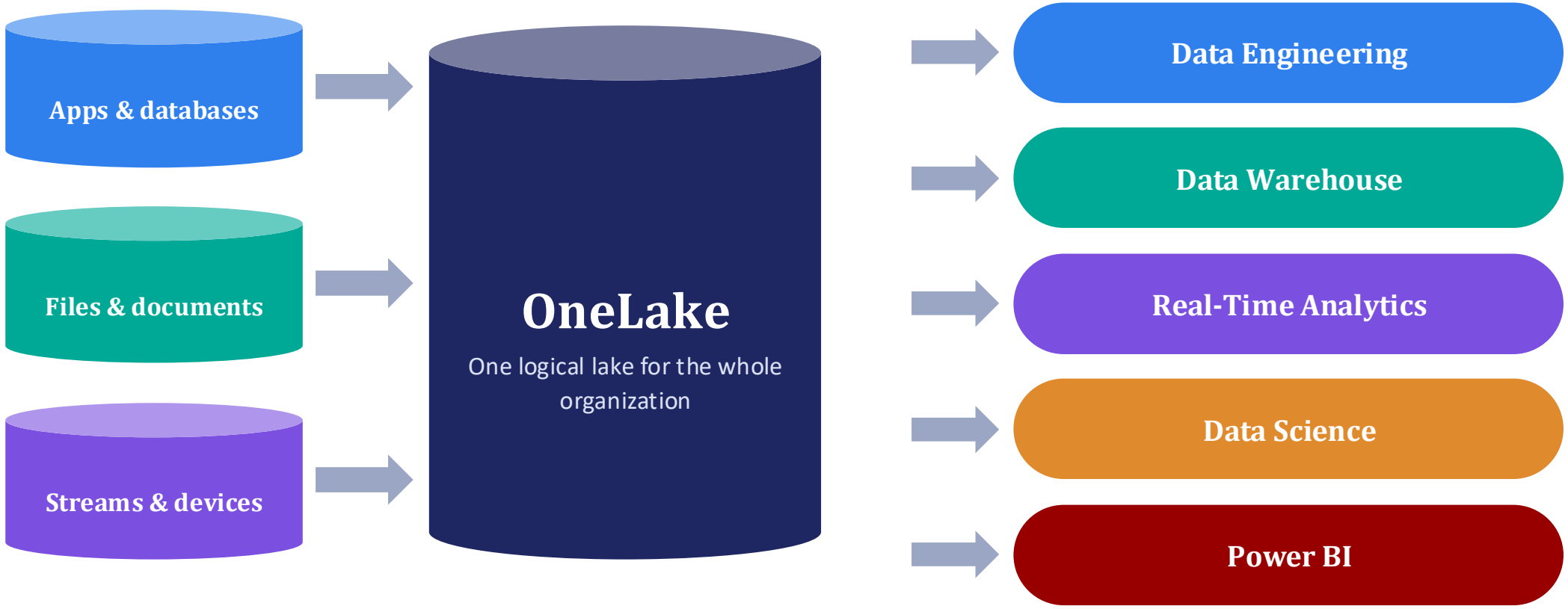
Images, Audio & Video

Rich media from apps and devices

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OneLake & the Fabric Data Estate

ONE HOME FOR DATA



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OneLake: One Home for Data

WHY IT MATTERS

One Lake

one logical lake

A single, unified data lake for the whole organization.

One Copy

no needless duplicates

Use the same data across engines without copying it.

One Security

consistent control

Apply governance and access rules in one place.

OneLake is often described as the OneDrive for data.

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Basic Data Preparation & Cleansing

GETTING DATA AI-READY

Collect
Bring data together from source systems

1

2

Clean
Remove duplicates, errors and missing values

Transform
Reshape and standardize into usable form

3

4

Validate
Confirm accuracy before it reaches AI

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Before and After Cleaning

SEE THE DIFFERENCE

Before cleaning

- × Duplicate customer records
- × Blank or missing values
- × Mixed date formats
- × Typos in categories

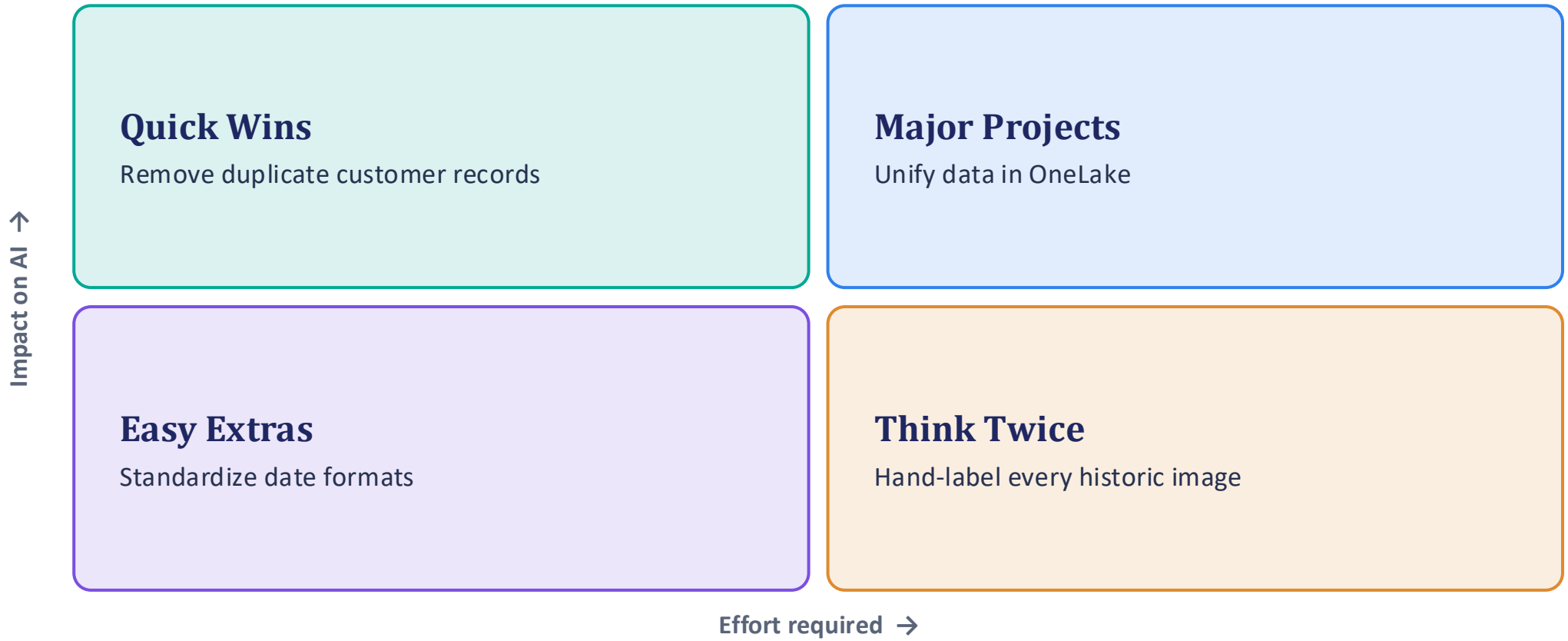
After cleaning

- ✓ One record per customer
- ✓ Gaps filled or flagged
- ✓ One standard format
- ✓ Consistent categories

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Where to Start: Prioritizing Data Work

IMPACT VS EFFORT



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Splitting Data for AI

TRAIN, VALIDATE, TEST



Illustrative split; common ratios vary

Training

Teaches the model patterns

Validation

Tunes settings during development

Test

Measures real-world performance

Never let test data leak into training, or results will look better than reality.

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Preparing Documents for AI

FOR RAG & SEARCH

CHUNK

Split long files into focused passages the model can retrieve

TAG

Add metadata like source, owner, date and sensitivity

SECURE

Keep access controls so people only retrieve what they may see

Well-prepared content makes retrieval and answers far more accurate.

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From Vague to Explicit Data Requests

BE EXPLICIT ABOUT DATA

Vague

"Give me the customer data."

Explicit

"Export active customers from the CRM for the last 12 months, remove duplicates, mask email addresses, as a CSV file."

Source | **Scope** | **Cleaning** | **Privacy** | **Format**

Specific requests produce data that is ready for AI, with fewer surprises.

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Governance & Handling Data Responsibly

PROTECTING WHAT MATTERS



Classify data by sensitivity level before it's used



Encrypt and control access to sensitive data



Comply with privacy regulations and internal policy



Maintain visibility into where data lives and flows

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Protecting Sensitive Data

TECHNIQUES IN ACTION

Masking

Hide values but keep the format

```
name:   J*** S****
email:  ****@contoso.com
```

Pseudonymize

Swap identifiers for tokens

```
customer_id: 48213   ->   C-9A7F
lookup table stored separately
```

Anonymize

Generalize so no one is identifiable

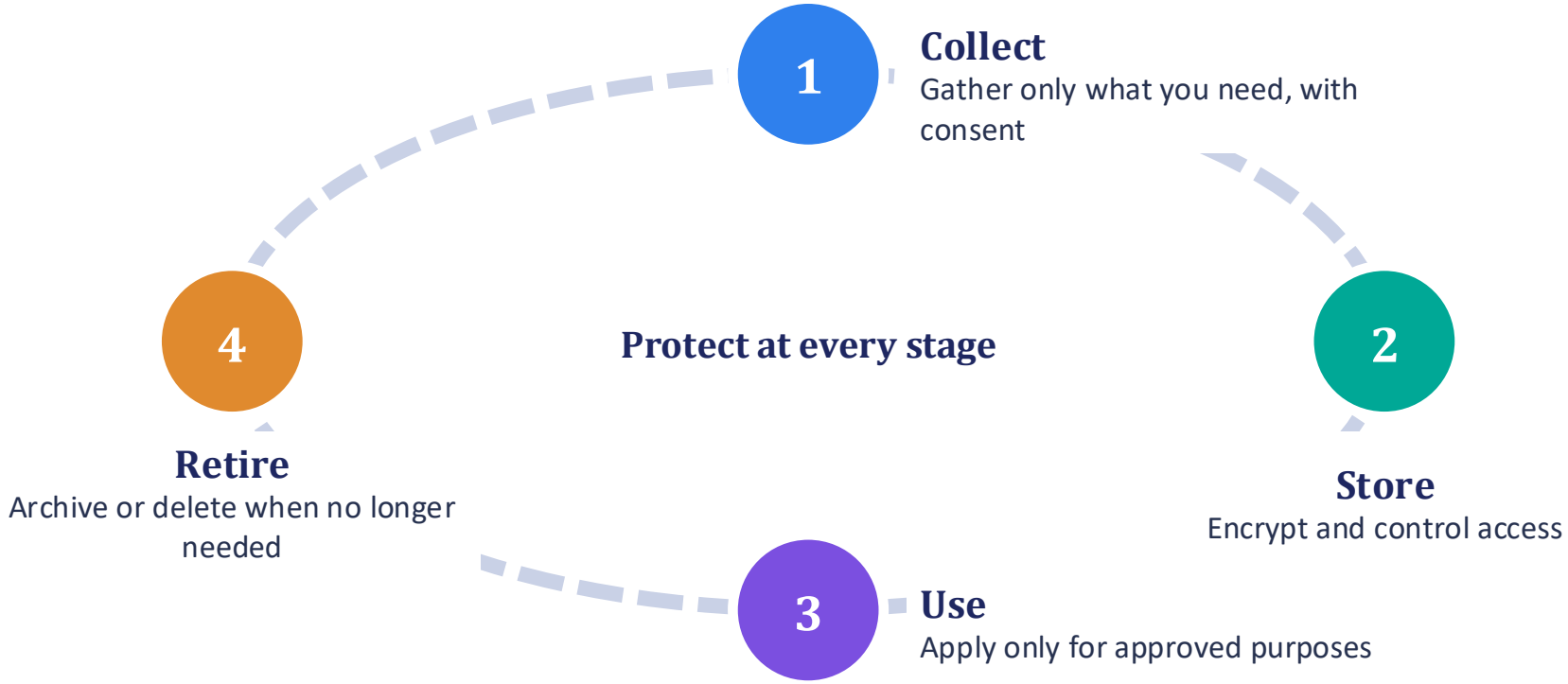
```
age:   34        ->   30-39
city:  Austin    ->   Texas
```

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The Data Lifecycle

PROTECT AT EVERY STAGE

Responsible handling does not end once data is collected.



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Hands-On: Audit a Dataset

TRY IT YOURSELF

⏱ 10 min

Your dataset

Pick a spreadsheet or table you use at work.

Quality checklist

- ☐ No duplicate rows
- ☐ Required fields filled
- ☐ Consistent formats
- ☐ Values look accurate
- ☐ Sensitive fields flagged

Note what you would fix first, and why.

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Data Fundamentals Recap

WRAP-UP

01	**Quality**	AI outcomes are only as good as their data
02	**Shapes**	Structured is organized; unstructured needs processing
03	**Sources**	Databases, apps, files and rich media feed AI
04	**OneLake**	Unifies Fabric engineering, warehouse and analytics
05	**Preparation**	Collect, clean, transform and validate first
06	**Governance**	Classify, control access and comply

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THANK YOU